DERIVED COMPLETE MODULES FORM A WEAK SERRE SUBCATEGORY AND THE DERIVED COMPLETION IDENTIFIES THE CORRESPONDING DERIVED SUBCATEGORY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The author writes "the long exact sequence of $\Ext$'s" in the counted proof span; the possessive construction is reproduced unchanged. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The Ext characterisation of derived completeness: vanishing of Ext^n_A(A_f, -) for all n and all f in the ideal. invoked at proof lines 26079-26085. Is the criterion used in both directions of the argument.
- The characterisation of weak Serre subcategories. invoked at proof lines 26091-26092. Packages the two verified axioms into the stated conclusion.

Source reference. Original lemma 0754: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9118–9118. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 25850: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25850–25850. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let I be an ideal of a ring A.*

(1) *The derived complete A-modules form a weak Serre subcategory $\mathcal{C}$ of Mod_A.*
(2) *$D_{\mathcal{C}}(A) \subset D(A)$ is the full subcategory of derived complete objects.*

Proof. Part (2) is immediate from Lemma 25850 and the definitions. For part (1), suppose that $M \to N$ is a map of derived complete modules. Denote $K = (M \to N)$ the corresponding object of $D(A)$. Pick $f \in I$. Then $\mathrm{Ext}^n_A(A_f, K)$ is zero for all

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n because $\mathrm{Ext}^n_A(A_f, M)$ and $\mathrm{Ext}^n_A(A_f, N)$ are zero for all n. Hence K is derived complete. By (2) we see that $\mathrm{Ker}(M \to N)$ and $\mathrm{Coker}(M \to N)$ are objects of $\mathcal{C}$. Finally, suppose that $0 \to M_1 \to M_2 \to M_3 \to 0$ is a short exact sequence of A-modules and M_1, M_3 are derived complete. Then it follows from the long exact sequence of Ext's that M_2 is derived complete. Thus $\mathcal{C}$ is a weak Serre subcategory by Homology, Lemma 0754. $\square$